

Worksheet: Introduction to Python Game Programming

Name: _____ Class: _____ (_____)

Section 1: Scaffolding – Understanding Modules

Pay attention to the presentation and fill in the blanks.

1. Python modules are programs written by other programmers that we can _____ in our own programs.
 2. The **random** module is a standard module used for things like generating a _____ number.
 3. To generate a random number between 1 and 32, we use the function: randint(_____, _____).
 4. In this course, we will use two special modules for game development: **pygame** and _____.
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Section 2: Discovery – File Organization

Look at the files on your computer and the presentation. Map out where everything goes.

To make the game work, you must arrange your files in specific folders:

- **Main Folder:** Contains your Python code file named _____.
- _____ **Folder:** Contains image files like monster.____.
- _____ **Folder:** Contains audio files like monsterdie.____.

Discovery Question: Why do you think PGZero requires images and sounds to be in specifically named folders?

Section 3: Game Logic & Functions

Complete the sentences using the technical terms from the slides.

1. **Basic Config:** To set the size of the game window, we use the variables _____ and _____.
 2. **Actors:** In a game, characters or objects are called **Actors** or _____.
 3. **Redrawing:** The _____ function is called by PGZero whenever the game window needs to be refreshed, such as after a mouse click.
 4. **Collision:** To check if a user clicked on a monster, we use the command monster1. _____(pos).
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Section 4: Discovery – Variable Scope

Read the section on "Variables and Functions" and answer the following.

1. A variable defined **inside** a function is called a _____ variable and is destroyed once the function finishes.
2. A variable defined **outside** a function is called a _____ variable.
3. To change a variable that exists outside your function (like `living_monsters`), you must use the keyword _____ inside your function first.

Experiment: What happens to the game if you forget to use the `global` keyword when updating the monster count?

Section 5: Modification Challenge

Play around with the code and try to "discover" how to change the game.

- **Change the sprite:** Find a new image. What is the recommended size for the monster sprite? _____.
- **Change the sound:** What file format must your new sound effects use? _____.
- **Game Goal:** What happens in the code when the number of `living_monsters` reaches 0? _____.